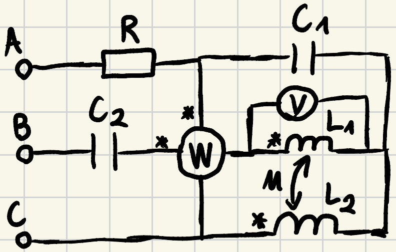
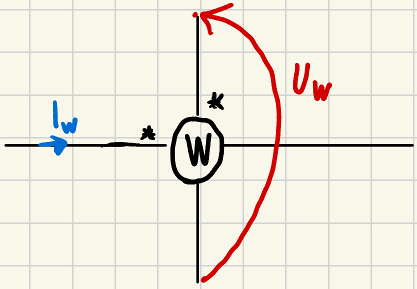
WYKŁAD 8

Przykład

Obliczyć prądy w dwadzie $\lambda - \lambda$ oraz wskazanie woltomierza i watomierza





$$P = Re[U_W \cdot I_W^*]$$

$X_{L_1} = 30\ \Omega$ $\quad R = 10\ \Omega$

$X_{L_2} = 10\ \Omega$ $\quad |U_F| = 200V$

$X_M = 10\ \Omega$

$X_{C1} = X_{C2} = 10\ \Omega$

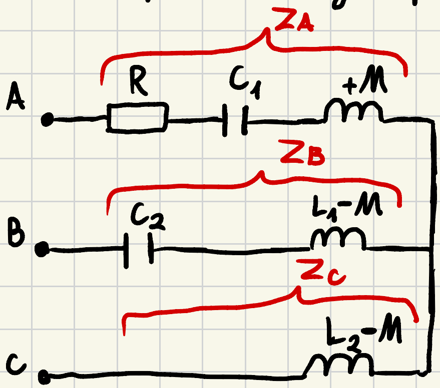
napięcia fazowe generatora

$$E_A = 200$$

$$E_B = 200 e^{-j120°} = 200(\cos 120° - j \sin 120°) = -100 - j173$$

$$E_C = 200 e^{+j120°} = 200(\cos 120° + j \sin 120°) = -100 + j173$$

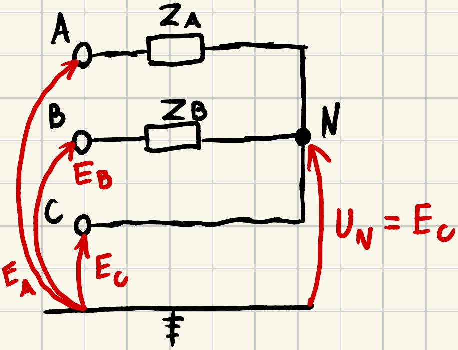
Obwód po eliminacji sprzężeń do obliczeń prądów



$$Z_A = R + jX_M - jX_{C1} = 10 + j10 - j10 = 10$$

$$Z_B = -jX_{C2} + jX_{L1} - jX_M = -j10 + j30 - j10 = j10$$

$$Z_C = jX_{L2} - jX_M = j10 - j10 = 0 \rightarrow Y_C = \infty$$



$$U_N = \frac{E_A Y_A + E_B Y_B + E_C Y_C}{Y_A + Y_B + Y_C} = \frac{E_A Y_A + E_B Y_B + E_C \cdot \infty}{Y_A + Y_B + \infty} = E_C$$

$$U_N = E_C = -100 + j173$$

$$\left.\begin{aligned}
U_A &= E_A - U_N = 300 - j173 \\
U_B &= E_B - U_N = -2 \cdot 173j = -j346 \\
U_C &= E_C - U_N = 0
\end{aligned}\right\}\ \text{napięcia fazowe odbiornika}$$

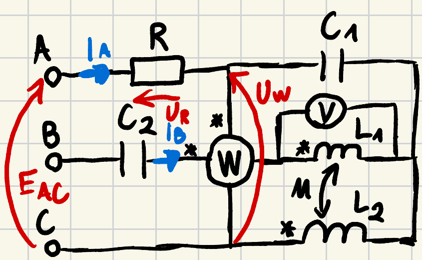
$$I_A = \frac{U_A}{Z_A} = \frac{300 - j173}{10} = 30 - j17{,}3$$

$$I_B = \frac{U_B}{Z_B} = \frac{-j346}{j10} = -34{,}6$$

$$I_C = \frac{U_C}{Z_C} = \frac{0}{0} = ? \rightarrow I_A + I_B + I_C = 0$$

$$I_C = -I_A - I_B = 4{,}6 + j17{,}3$$

Watomierz



$$P_W = Re[U_W \cdot I_W^*] = Re[0 \cdot 0] = 0$$

$$P_W = 0$$

$$I_N = I_B = -34{,}6$$

$$U_W = E_{AC} - U_R = E_{AC} - RI_A = E_A - E_C - RI_A$$

$$U_W = 200 + 100 - j173 - 10(30 - j17{,}3)$$

$$U_W = 0$$

Napięcie na L_1

$$U_{L1} = jX_{L1} \cdot I_B + jX_M \cdot I_C$$

$$U_{L1} = j30 \cdot (-34{,}6) + j10(4{,}6 + j17{,}3)$$

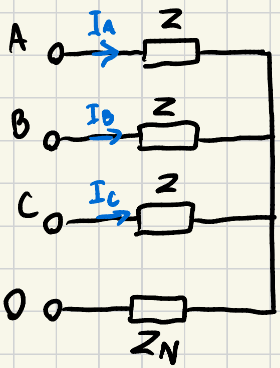
$$U_{L1} = -173 - j992$$

$$|U_{L1}| = U_{volt} = \sqrt{173^2 + 992^2} = 1007\,[V]$$

Przypadek symetrycznego obciążenia (λ - λ)

$$Z_A = Z_B = Z_C = Z$$

$$E_B = E_A \cdot e^{-j120^\circ} \qquad E_C = E_A \cdot e^{j120^\circ}$$



$$U_N = \frac{E_A \cdot Y + E_B \cdot Y + E_C \cdot Y}{Y + Y + Y + Y_N} = \frac{Y(\overset{=0}{E_A + E_B + E_C})}{3Y + Y_N} = 0$$

$$U_A = E_A - U_N = E_A - 0 = E_A$$

$$U_B = E_B$$

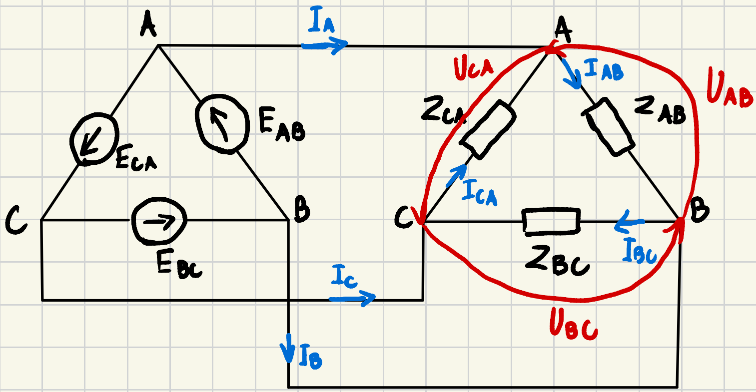
$$U_C = E_C$$

$$I_A = \frac{E_A}{Z_A} = \frac{1}{Z} \cdot E_A$$

$$I_B = \frac{E_B}{Z_B} = \frac{1}{Z} \cdot E_B = \frac{1}{Z} \cdot E_A \cdot e^{-j120^\circ} = I_A \cdot e^{-j120^\circ}$$

$$I_C = \frac{E_C}{Z_C} = \frac{1}{Z} \cdot E_C = I_A \cdot e^{j120^\circ}$$

Połączenie $\Delta - \Delta$



$$\begin{cases} U_{AB} = E_{AB} \\ U_{BC} = E_{BC} \\ U_{CA} = U_{CA} \end{cases} \longrightarrow \begin{cases} I_{AB} = \dfrac{E_{AB}}{Z_{AB}} \\ I_{BC} = \dfrac{E_{BC}}{Z_{BC}} \\ I_{CA} = \dfrac{E_{CA}}{Z_{CA}} \end{cases} \longrightarrow \begin{array}{l} I_A = I_{AB} - I_{CA} \\ I_B = I_{BC} - I_{AB} \\ I_C = I_{CA} - I_{BC} \end{array}$$

Przykład:

$Z_{AB} = R = 10\,\Omega$

$Z_{BC} = jX_L = j10\,\Omega$

$Z_{CA} = -jX_C = -j10\,\Omega$

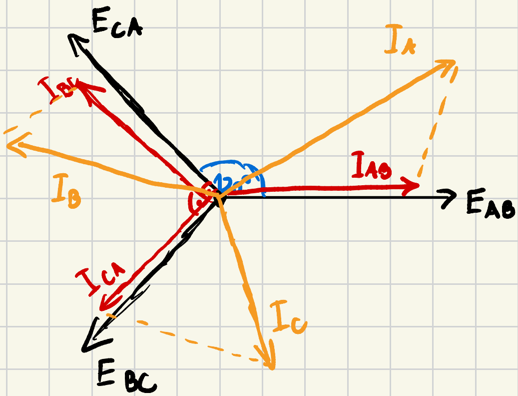
$|U_F| = 200V$

$$E_{AB} = 200\sqrt{3} \rightarrow I_{AB} = \frac{200\sqrt{3}}{10} = 20\sqrt{3} \rightarrow I_A = I_{AB} - I_{CA}$$

$$E_{BC} = 200\sqrt{3}\,e^{-j120^\circ} \rightarrow I_{BC} = \frac{200\sqrt{3}\,e^{-j120^\circ}}{10e^{j90^\circ}} = 20\sqrt{3}\,e^{-j210^\circ} \rightarrow I_B = I_{BC} - I_{AB}$$

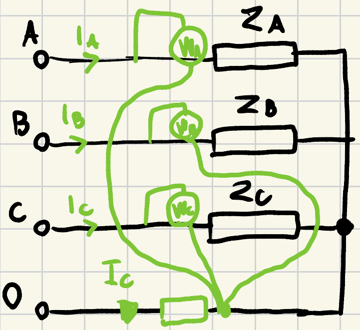
$$E_{CA} = 200\sqrt{3}\,e^{j120^\circ} \rightarrow I_{CA} = \frac{200\sqrt{3}\,e^{j120^\circ}}{10e^{-j90^\circ}} = 200\sqrt{3}\,e^{j210^\circ} \rightarrow I_C = I_{AC} - I_{BC}$$

Wykres wektorowy



Pomiary mocy w obwodach 3-fazowych

1. Moc w układzie $\lambda - \lambda$ czteroprzewodowym



Moc odbiornika

$$P_{3f} = P_A + P_B + P_C$$

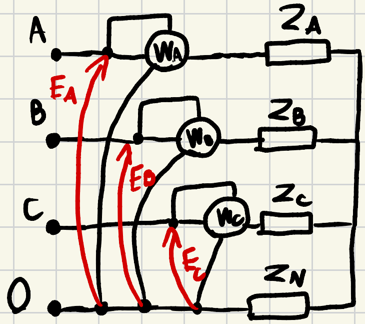
Moc odbiornika chwilowa

$$p(t) = p_A(t) + p_B(t) + p_C(t)$$

$$p(t) = u_A(t) \cdot i_A(t) + u_B \cdot i_B(t) + u_C \cdot i_C(t)$$

Moc czynna

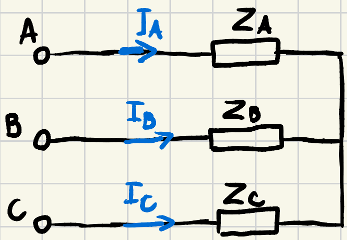
$$P = \frac{1}{T} \int_{t_0}^{t_0+T} p(t)\, dt = \frac{1}{T} \Big[\underbrace{\int_{t_0}^{t_0+T} u_A(t) \cdot i_A(t)\, dt}_{|U_A| \cdot |I_A| \cdot \cos\varphi_A} + \underbrace{\int_{t_0}^{t_0+T} u_B(t) \cdot i_B(t)\, dt}_{|U_B| \cdot |I_B| \cdot \cos\varphi_B} + \underbrace{\int_{t_0}^{t_0+T} u_C(t) \cdot i_C(t)\, dt}_{|U_C| \cdot |I_C| \cdot \cos\varphi_C} \Big]$$



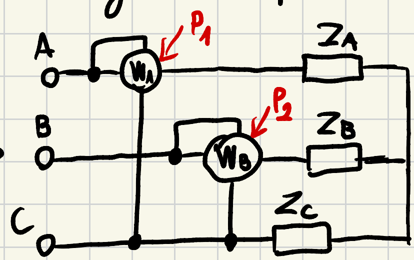
Moc czynna generatora

$$P_{3f} = |E_A||I_A|\cdot\cos\varphi_A + |E_B||I_B|\cos\varphi_B + |E_C|\cdot|I_C|\cdot\cos\varphi_C$$

Obwód 3-fazowy 3-przewodowy (brak przewodu neutralnego)



$\rightarrow$



++ Pokemon Reference ++

Układ Arona

$$P = P_1 + P_2$$

$$I_A + I_B + I_C = 0 \qquad i_A(t) + i_B(t) + i_C(t) = 0 \rightarrow i_C(t) = -i_A(t) - i_B(t)$$

$$p(t) = u_A(t)\cdot i_A(t) + u_B(t)\cdot i_B(t) + u_C(t)\cdot i_C(t)$$

$$p(t) = u_A(t)\cdot i_A(t) + u_B(t)\cdot i_B(t) + u_C(t)\cdot(-i_A(t) - i_B(t))$$

$$p(t) = (u_A(t) - u_C(t))\cdot i_A(t) + (u_B(t) - u_C(t))\cdot i_B(t)$$

$\downarrow$ wartość zespolona

$$P = |U_{AC}|\cdot|I_A|\cdot\cos\varphi_{U_{AC},I_A} + |U_{BC}|\cdot|I_B|\cdot\cos\varphi_{U_{BC},I_B}$$

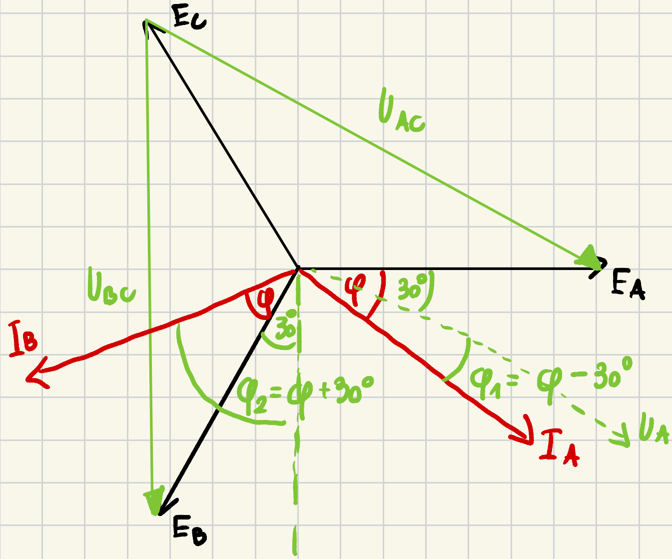
Układ 3-przewodowy symetryczny $Z_A = Z_B = Z_C = Z$

$E_A = U_A$

$E_B = U_B$

$E_C = U_C$

$|I_A| = |I_B| = |I_C| = |I_f|$



$$P_1 = \sqrt{3} \cdot |U_f| \cdot |I_f| \cdot \cos(\varphi - 30°)$$

$$P_2 = \sqrt{3} \cdot |U_f| \cdot |I_f| \cdot \cos(\varphi + 30°)$$

$$P = P_1 + P_2 = \sqrt{3}\, |U_f| \cdot |I_f| \left(\cos(\varphi - 30°) + \cos(\varphi + 30°) \right)$$

$P = P_1 + P_2$

$$P = \sqrt{3}\, |U_f| \cdot |I_f| \left(\cos\varphi \cos 30° + \cancel{\sin\varphi\, 30°} + \cos\varphi \cos 30° - \cancel{\sin\varphi \sin 30°} \right)$$

$$P = \sqrt{3}\, |U_f| \cdot |I_f| \cdot 2\cos\varphi \cdot \frac{\sqrt{3}}{2} = \underbrace{3 \cdot |U_f| \cdot |I_f| \cdot \cos\varphi}_{P_{1\,fazy}}$$

$P_1 - P_2$

$$P_1 - P_2 = \sqrt{3} \cdot |U_f| \cdot |I_f| \cdot 2\sin\varphi \sin 30° = \sqrt{3}\, |U_f|\, |I_f| \cdot \sin\varphi = \sqrt{3} \cdot Q_{1\,fazy}$$

Pełna moc bierna 3 faz

$$Q = 3 \cdot Q_{1faz} = 3 \cdot |U_f| \cdot |I_f| \cdot \sin\varphi = 3 \cdot \frac{P_1 - P_2}{\sqrt{3}} = \sqrt{3}(P_1 - P_2)$$

Moc czynna 3 faz

$$P = P_1 + P_2$$

Moc bierna 3 faz

$$Q = \sqrt{3}(P_1 - P_2)$$

Kąt fazowy φ

$$\varphi = \operatorname{arctg}\frac{Q}{P} = \operatorname{arctg}\left[\frac{\sqrt{3}(P_1 - P_2)}{P_1 + P_2}\right]$$